

Philip (Phi) Bui

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EDUCATION

Worcester Polytechnic Institute, B.S. in Computer Science, Dean's List, [Upsilon Pi Epsilon](#) Aug 2021 - Dec 2023
Tufts University, M.S. in Computer Science: Human-Robot Interaction, [Honos Civicus Society](#) Sep 2024 - May 2025
New York University, Ph.D. in Urban Systems, [Tandon School of Engineering Fellow](#) Sep 2026 - Present

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer, Cyvl Aug 2025 - Aug 2026
• Developed digital twin technologies for cities to improve civil infrastructure and urban planning. $0 \rightarrow 1 \rightarrow \infty$.

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• Developed digital twin technologies for cities to improve civil infrastructure and urban planning. $0 \rightarrow 1 \rightarrow \infty$.
• Developed geospatially-aware AI municipal document ingestion pipelines (won NYC DEP ETL Challenge).
• Shipped dozens of features for thousands of civic executives via LiDAR/imagery analysis at petabyte scale
• Partnered with City of Boston on MCP servers giving non-technical staff accessible city-sensing data access.

Research Intern, AI Futures Collab, WPI May - Sep 2025
• Investigated the phenomenology and affordances behind human/AI interfaces.

Swarm Robotics AI Research Intern, Air Force Research Laboratory May - Aug 2025
• Developed specialized, heterogeneous multi-agent reinforcement learning algorithms for swarm systems.

Graduate Research Assistant, Human Interaction Lab, Tufts University Jan - May 2025
• Refactored GailBot, a Dockerized Python app for AI transcription and speech analysis with multiple engines.
• Deployed NextJS via Vercel, integrating TypeScript RESTful APIs & AWS S3; [Presented @ ICCA 2026](#).

Full-Stack Software Engineer, enLabel Jan 2024 - Jan 2025
• Architected a React TypeScript frontend with ASP.NET RESTful API integration & end-to-end regression tools.
• Contributed to technical planning and architecture meetings with cross-functional stakeholders & clients.
• Led design specification, timeline management, and strategic planning of engineering deliverables & docs.

Website Developer, ProsperOn Jun 2023 - Feb 2024
• Built React/TypeScript financial simulation web apps & Node.js RESTful APIs, modernizing legacy code.

Operations Intern, Amazon Jun - Aug 2023
• Developed an internal, mapping application, reducing costs by >\$10k (95%) per trained employee.
• Built a Node.js server for data retrieval via Amazon's internal database, located and monitored >7,000 items.

PROJECTS

State-Wise Constrained Policy Shaping: Runtime Behavior Steering for Safe RL, Tufts University Jan 2024 - 2025
• Developed SCPS: a post-training supervisor for RL in HighwayEnv that enforces safety constraints at runtime.
• Reduced collisions by 97% in-distribution and 99% zero-shot; [Presented @ AAAI-26: AIGOV in Singapore](#).

LLM Enhanced Captioning for Neurodivergent Users, Tufts University Jan - May 2025
• Developed a browser extension using Mistral-7B for enriched captions; <200 ms latency & >95% accuracy.

Computer Vision-Guided Sphero Swarm Navigation Framework, Tufts University Sep - Dec 2024
• Developed collision-free PRM path planning & CV MCL SLAM for swarms via BLE & WebSocket microservices.

AI Platform for Low-Income Housing Development for City of Worcester, MA, WPI MQP Aug - Dec 2023
• Developed an AI-driven due diligence and GIS analysis platform; Owned by City of Worcester, MA.

COMMUNITY EXPERIENCE

Co-Founder & Chief Logistics Officer, Untitled LLC Dec 2023 - Present
Community Events Coordinator, WPI Powerlifting Jan 2023 - Dec 2023